



EDU TERIA

72nd BPSC

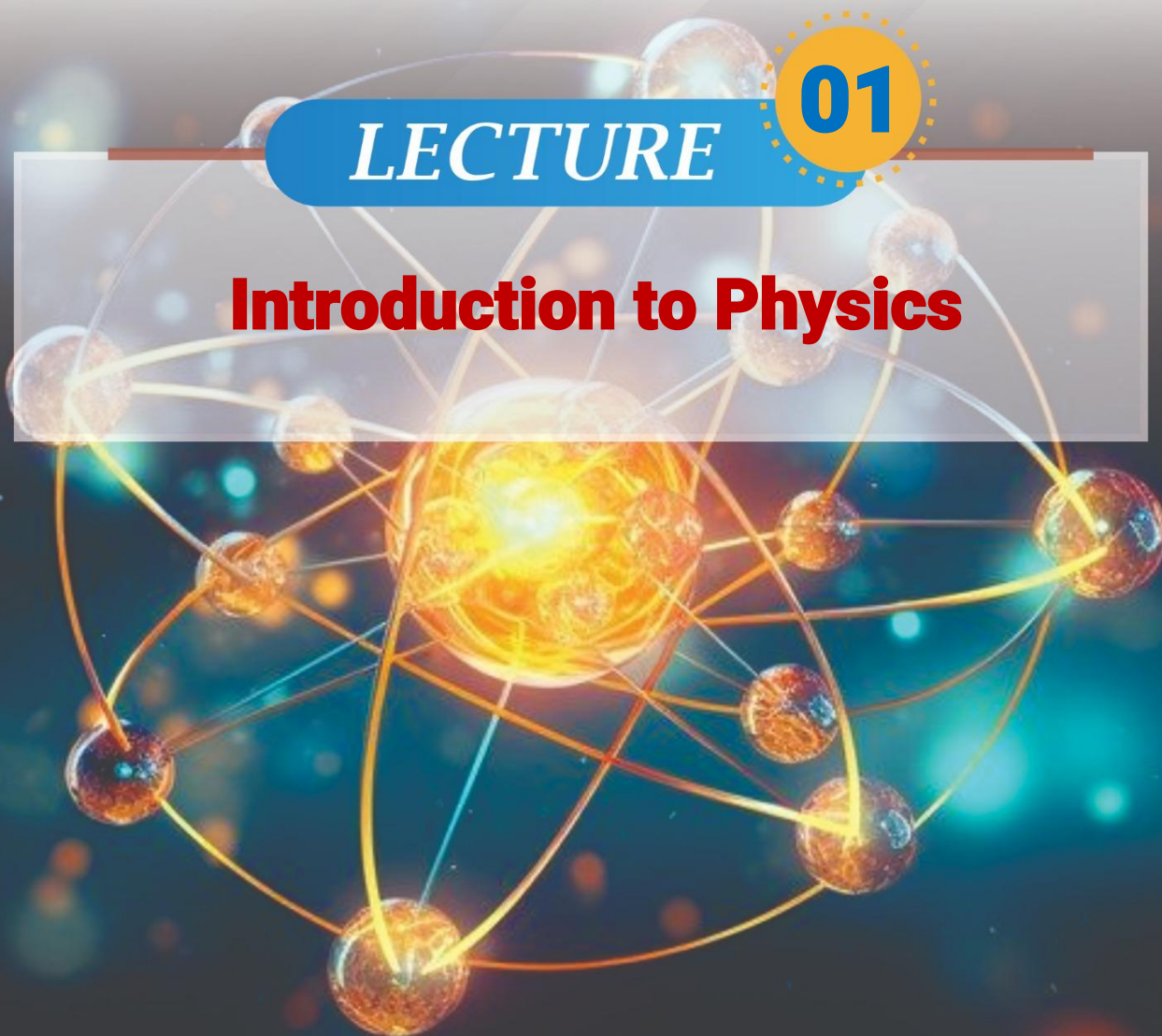
Prelims Online Batch

PHYSICS

LECTURE

01

Introduction to Physics



**Introduction to Physics****Class will be divided into following schedule:****Introduction to Physics**

Light (1) - Nature and properties of light, reflection of light and mirrors, types of mirrors and their uses.

Light (2)- Refraction of light, total internal reflection, lenses and their types, uses of lenses and correction of eye defects.

Light (3) Scattering of light, dispersion of light, polarization, interference and their examples.

Waves - Types of waves, mechanical waves and their properties, non-mechanical waves (electromagnetic waves), their properties and uses.

Sound - waves Speed of sound waves, echo
Mach number, shock waves, Doppler effect.

Heat (1) - Heat and temperature, thermometer and its scale, methods of heat conduction, thermal expansion and its examples.

Heat (2) - Anomalous behavior of water, specific heat, latent heat, evaporation and humidity (relative humidity), precipitation.

Motion - Basic elements of motion - distance, displacement, speed, velocity, and acceleration, force and types of force, Newton's laws of motion.

Types of - Motion Such as linear motion, uniform circular motion, rotational motion, projectile motion, periodic motion, oscillatory motion, simple harmonic motion, etc.

Universal - Gravitation Gravitational force and gravity,

Newton's law of gravitation, gravitational Acceleration, changes in the value of g , orbital velocity, escape velocity, Kepler's planetary laws

Mechanical- Cohesive and cohesive forces, fluid

properties of fluids pressure and Pascal's law, surface tension and capillarity, streamline flow and Bernoulli's theorem, buoyancy and Archimedes' principle

Electricity (1) Charge, electric field, electric field intensity, electric potential, potential difference, electric capacitance and capacitors, electric current, Ohm's law and resistance

Electricity (1) Electric power, commercial units (kilowatt-hour units), electromagnetic induction, electric motors and generators

Magnetism- Magnetic field, magnetic flux, magnetic permeability, magnetic susceptibility, types of magnetic materials, geomagnetism





Light

Optics: Branch of physics which deals with behaviour of light.

Propagation of Light

Light travels along straight line path in a certain medium or in vacuum. The path of light changes only where the medium changes. We call this rectilinear (straight-line) propagation of light. A bundle of light rays is called a beam of light.

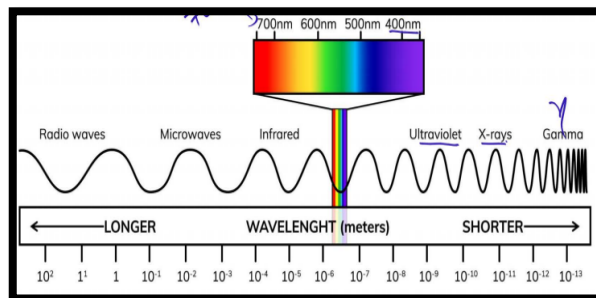
Apart from vacuum and gases, light can travel through some liquids and solids as well. A medium in which light can travel without attenuation over large distances is called a transparent medium. Water, glycerine, glass and clear plastics are transparent. A medium in which light cannot travel is called opaque. Wood, metals, bricks, etc., are opaque. In materials like oil, light can travel some distance, but its intensity reduces rapidly. Such materials are called translucent.

Two components of optics —

- Ray optics** – Wave nature of light.
 - Wave optics** – Wave nature of light
- Light has dual nature particle as well as wave:
 - Wave nature (interference, diffraction)
 - Particle nature (photoelectric effect)
 - Light is considered as electromagnetic wave, [Electromagnetic wave means, it does not require medium to travel].
 - Speed of light In vacuum – 3×10^8 m/s
 - Water – 2.25×10^8 m/s
 - Glass – 2×10^8 m/s
 - Light is made up of visible spectrum whose wavelength lies between $3800\text{\AA}$ to $7500\text{\AA}$.

Light

Visible Light



Visible Spectrum

V	I	B	G	Y	O	R
Violet $\lambda = 4000\text{\AA}$	Indigo	Blue	Green	Yellow	Orange	Red $\lambda = 7000\text{\AA}$

H = Planck constant

C = Velocity of light in vacuum

(i) Least sensitive colour: violet & red

(ii) Most sensitive colour for human eye → yellow

$$\downarrow E = h\nu \downarrow$$

(iii) Soothing colour → Green

Formula

$$C = \lambda \times \nu$$

c = speed of light

ν = frequency

λ = wavelength

Frequency $\uparrow \Rightarrow$ Wavelength $\downarrow$ (inverse relation)

Scattering of light $\propto \frac{1}{\lambda^4}$ to spread visible spectrum.

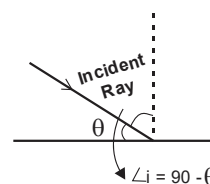
Reflection of Light

Reflection: Bouncing back of light from a surface.

Laws of Reflection

- Incident ray, reflected ray & normal lie in same plane
- Angle of incidence = angle of reflection

$$\angle i = \angle r$$



**Reflection of Light**

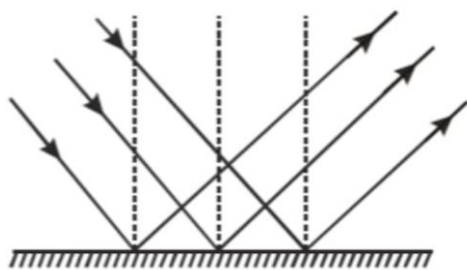
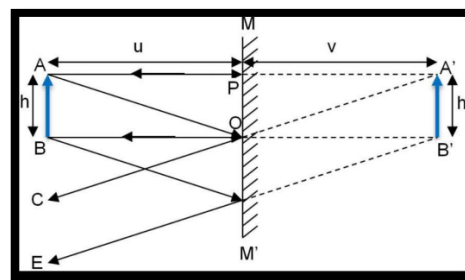
When light rays strike the boundary of two media such as air and glass, a part of light bounces back into the same medium. This phenomenon of light is called Reflection of light.

(i) Regular / Specular reflection :

When reflection takes place from a perfect plane surface

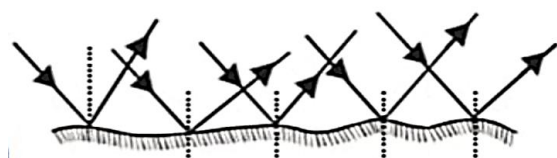
then rays remain parallel after reflection .

It is called Regular reflection.

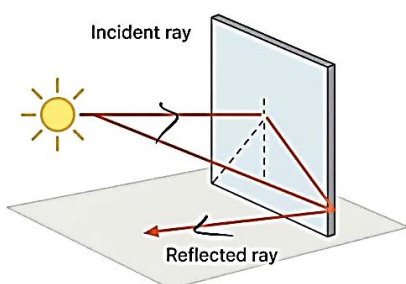
Reflection with plane mirror**(ii) Irregular / Diffused reflection**

When the surface is rough, light is reflected from the bits of its plane surfaces in different directions.

This is called Irregular reflection. This process enables us to see an object from any position.



Reflection of light from mirror

**Reflection**